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Inductive inference of unbounded unions of pattern languages from positive data

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Abstract

A pattern is a string consisting of constant symbols and variables. The language of a pattern is the set of constant strings obtained by substituting nonempty constant strings for variables in the pattern. In this paper we consider the inductive inference of unbounded unions of pattern languages from positive data. For any fixed k, the class of unions of at most k pattern languages is already shown to be inferable from positive data. The class of unbounded unions is not inferable, because any pattern with at least one variable defines an infinite language and any constant string defines a singleton set consisting of itself, and therefore, the class of unions becomes *super-finite*, that is, it contains all the finite languages and at least one infinite language. In this paper, we consider several restrictions on patterns to investigate the inferability of unbounded unions of their languages from positive data. A *proper* pattern contains at least one variable. A *regular* pattern contains at most one occurrence of every variable. Although the class of unbounded unions of proper pattern languages is not super-finite, it is shown not to be inferable from positive data, even if patterns are restricted to be regular or one-variable. When regular patterns are restricted to be of the form "xwy", where x and y are variables and w is a constant string, the class of unbounded unions is shown to be inferable from positive data. When regular patterns do not contain more than l consecutive occurrences of constant symbols, for some fixed l, the class of unbounded unions is shown to be inferable from positive data. Extended pattern languages, where substitutions of the empty string for some variables are allowed, are also considered from the viewpoint of relationship to the inferability of unbounded unions of non-extended pattern languages. © 2000 Elsevier Science B.V. All rights reserved.

1. Introduction

Inductive inference is formalized as an infinite process of guessing hypothesis based on successively given concrete examples from an unknown concept. When a sequence of hypotheses produced by a process eventually converges to a correct representation

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of the target concept, the process is said to *identify* the concept *in the limit* [9]. Since for any concept Γ, there exists at least one trivial process that identifies Γ, the most important question is whether an inference process carried out by a fixed program can identify any member of a class of concepts.

Inductive inference of languages from positive data is an inductive inference when concepts to be inferred are languages and their examples are taken from the inside of the target language. A class of languages is said to be *super-finite* when it contains all the finite languages and at least one infinite language. Gold [9] showed that any super-finite class is *not* inferable from positive data. However, Angluin [1, 2] pointed out that there remains not a small possibility in inductive inference from positive data and introduced the class of *pattern languages*, which is inferable from positive data. She showed a necessary and sufficient condition for a class to be inferable from positive data. From the condition, we know that every language L in a class inferable from positive data should have a *finite tell-tale*, that is a finite subset of L that is not contained in any proper subset of L in the class. For instance, any infinite language in a super-finite class has no finite tell-tale.

A pattern is a string consisting of constant symbols from a finite alphabet Σ and variables from a countable alphabet X. The language $L(p)$ of a pattern p is the set of constant strings obtained by substituting nonempty constant strings for variables in p. A regular pattern is a pattern containing at most one occurrence of each variable. For example, when $\Sigma = \{a, b\}$ and $X = \{x, y, \ldots, \}$, $p = axby$ is a regular pattern but $q = axbx$ is not a regular pattern, and their languages are $L(p) = \{aubv \mid u, v \in \Sigma+\}$ and $L(q) = \{aubu \mid u \in \Sigma+\}$, respectively. Clearly, any regular pattern defines a regular language.

Although pattern languages are very simple, their inductive inference from positive data has many potential applications in practice. Shinohara [17, 18] showed that the class of regular pattern languages is efficiently inferable from positive data and its inductive inference can be utilized as a learning function of common form in a data entry system. For example, if we were entering bibliographic data, then we would type in the first few records:

```
$
Author: Angluin, D.
Title:  Inductive Inference of Formal Languages from Positive Data
Journal: Information and Control, 45, 117-135
Year:   1980
$
Author: Kruskal, J. B.
Title:  The theory of well-quasi-ordering: a frequently
        discovered concept.
Journal: Journal of Combinatorial Theory (A), 13, 297-305
Year:   1972
$
...
```

where "$" is a special symbol to delimit records. Then, the system would infer the form of records in a regular pattern

$$\texttt{Author:}\square\square x_1\texttt{Title:}\square\square\square x_2\texttt{Journal:}\square x_3\texttt{Year:}\square\square\square\square x_4$$

where "□" represents a blank symbol, and the system would emit the constant part, such as "`Author: `", and wait for us to type in the field corresponding to the variable part.

Nix [16] paid attention to routine work in text editing and realized a text transformation program synthesis from input/output examples based on inductive inference of pattern languages. For example, consider a transformation of a list baseball score like

```
Yankees 3, Orioles 1.
Brewers 12, Reds 5.
Dodgers 5, Braves 4.
...
```

to a database input format

```
GameScore[ winner 'Yankees'; loser 'Orioles'; scores [ 3,1 ] ];
GameScore[ winner 'Brewers'; loser 'Reds'; scores [ 12,5 ] ];
GameScore[ winner 'Dodgers'; loser 'Braves'; scores [ 5,4 ] ];
```

Such transformation is synthesized by the system from examples as a program for the form

$$x_1\square x_2,\ \square x_3\square x_4.$$
$$\Rightarrow \texttt{GameScore[}\square\texttt{winner}\square\,\texttt{'}x_1\texttt{'};\square\texttt{loser}\square\,\texttt{'}x_3\texttt{'};\square\texttt{scores}\square\,\texttt{[}\square x_2,\square x_4\square\texttt{]}\square\texttt{];}$$

For some applications, however, pattern languages are not directly used because of their simplicity. The class of pattern languages is not closed under union. Therefore, the class of their unions is richer than the class of pattern languages. Shinohara [19] showed that the class of unions of *two* pattern languages is inferable from positive data. This result is extended to the classes of unions of arbitrarily *fixed* number of pattern languages by Wright [24]. Arimura et al. [6, 7] showed an efficient inference algorithm for unions of regular pattern languages. Arikawa et al. applied learning algorithms, based on PAC learning model [23], to machine discovery of protein motifs from amino acid sequences using unions of regular pattern languages [4] and combination of decision trees with regular patterns [5]. Yamaguchi et al. [25] reported experimental results of learning algorithm of unions of regular patterns based on inductive inference from positive data.

In some practical applications, such as genome informatics, we do not know in advance an appropriate upper bound for the number of languages in unions. Thus, inductive inference of unbounded unions is very important in practical study as well as in theoretical one. Since any language consisting of just one string w is defined as a pattern language by w itself and any pattern containing at least one variable defines an infinite language, the class of unions of *unbounded* number of pattern languages

becomes super-finite and not inferable from positive data. Thus, to get inferable classes of unbounded unions, we have to restrict the forms of patterns or the number of pattern languages in unions. Several natural subclasses of patterns, such as one-variable patterns and regular patterns, have been introduced. Unless the subclass of patterns excludes some of patterns without variable, the class of unions remains super-finite. One of the main objectives of the present study is to reveal hot to restrict patterns or unions of pattern languages to establish unbounded unions that are inferable from positive data.

The *extended language* [18] of a pattern p, denoted by $L_\varepsilon(p)$, is the set of constant strings obtained by substituting *possibly empty* constant strings for variables in p. Extended pattern languages are also called erasing pattern languages [11]. Shinohara showed that the class of extended regular pattern languages is inferable from positive data [18]. However, the inferability of all the extended pattern languages from positive data is a very important open problem. The extended language $L_\varepsilon(p)$ is equal to a union of at most 2^m non-extended pattern languages where m is the number of variables in p. For instance, a string $w = aub \in L_\varepsilon(axby)$ is obtained by substituting a nonempty constant string u for variable x and erasing variable y, which means $w \in L(axb)$, and therefore, $L_\varepsilon(axby) = L(ab) \cup L(aby) \cup L(axb) \cup L(axby)$. By using this fact, Wright showed that the class of extended languages of m-variable patterns, which is a subclass of unions of at most 2^m non-extended pattern languages, is inferable from positive data [24]. A pattern is said to be *proper*, if it contains at least one variable. Note that all but one of the patterns in the right-hand side of the equation are proper. Thus, the inferability of extended pattern languages seems to be closely related to that of unbounded unions of proper pattern languages. Proper patterns have been introduced by Jantke and Lange to study case-based learning of pattern languages [10].

Since any proper pattern always defines an infinite language, the class of unbounded unions of proper pattern languages is not super-finite. The pattern x defines the language Σ^+ of all the nonempty strings. It is not hard to see that the singleton set $\{a\}$ is a finite tell-tale of $L(x)$ in the class of unions of proper pattern languages. Therefore, proper pattern languages are worth considering about the inferability of unions from positive data. We can find restrictions on patterns from practical applications. For example, Arikawa et al. [4] used regular patterns restricted to be of the forms "$x_1w_1x_2$", "$x_1w_1x_2w_2x_3$", or "$x_1w_1x_2w_2x_3w_3x_4$", where $x_!, x_2, x_3$ and x_4 are variables and w_1, w_2 and w_3 are non-empty constant strings. Their experimental results say that even these restricted regular patterns are practically useful to find classification rules for biosequences. A regular pattern $x_1w_1x_2$ represents the set of strings containing w_1 as its substring, and it is called a *substring pattern*. Note that the number of substring pattern languages is infinite, and the inferability of their unbounded unions is non-trivial.

In Section 3, we show that the class of unbounded unions of substring pattern languages is shown to be inferable from positive data. However, if we allow patterns of the form "$x_1w_1x_2w_2x_3$" additional to substring patterns, the class of unbounded unions is not inferable from positive data. Thus, it turns out that the class of unbounded

unions of proper pattern languages is not inferable from positive data, even if patterns are restricted to be regular as well as proper. We also show a negative results for proper one-variable pattern languages.

In Section 4, we consider restrictions on the length of *constant segments*, which are substrings of a pattern consisting of constant symbols. For example, a regular pattern $x_1abx_2aabx_3$ has two constant segments ab and aab. We consider more restricted regular patterns in which lengths of constant segments are at most l for some fixed l. Since the number of languages of substring patterns with bounded constant segments is finite, the class of unbounded union of their languages consists of finitely many languages, and therefore, it is obviously inferable from positive data. However, when we do not bound the number of constant segments in regular patterns, the class of unbounded unions of their languages becomes infinite and its inferability from positive data is non-trivial. By using Higman's Theorem [12] on well-quasi-ordering, we show the class of unbounded unions of regular patterns with constant segments of length at most l is inferable from positive data for arbitrarily fixed l.

2. Preliminaries

In this section, we give definitions and theorems concerning notions, such as, inductive inference from positive data, pattern languages, inductive inference of unions.

2.1. Inductive inference from positive data

First, we give one of the most important and essential mathematical models of learning from examples, called "inductive inference" or "identification in the limit".

A *word* over an alphabet Σ is a finite string of symbols taken from Σ. The *empty word* is a null string and denoted by ε. The length of a word w is denoted by $|w|$. Σ^*, $\Sigma^+, \Sigma^{=n}$ and $\Sigma^{\leqq n}$ denote the sets of all the words, nonempty words, words of length n, and words of length n or less, respectively. A subset of Σ^* is called a *language* over Σ. For a set S, $\#S$ denotes the number of elements in S. For a symbol $a \in \Sigma$ and a natural number n, a^n denotes the word of length n consisting only of a.

Let $I = \{1, 2, \ldots\}$ be the set of *indices*. A class of languages $\boldsymbol{L} = L_1, L_2, \ldots$ is said to be an *indexed family of recursive languages* if there exists a computable function $f : \Sigma^* \times I \to \{0, 1\}$ such that $f(w, i) = 1$ if $w \in L_i$ and $f(w, i) = 0$ otherwise. The index i of a language L_i can be considered as a description like a pattern or grammar which defines L_i. From here on, every class of languages is assumed to be an indexed family of recursive languages.

A *complete presentation* of a language $\boldsymbol{L}$ is an infinite sequence $(w_1, t_1), (w_2, t_2), \ldots$ such that t_i is 0 or 1, $\{w_i \mid i \in I, t_i = 1\} = L$, and $\{w_i \mid i \in I, t_i = 0\} = \Sigma^* - L$. A *positive presentation* of a nonempty languages $\boldsymbol{L}$ is an infinite sequence of $w_1, w_2, \ldots$ such that $\{w_i \mid i \in I\} = L$.

An *inductive inference machine* is an effective procedure that requests input from time to time and produces output from time to time. An output produced by an inductive

inference machine is called a *hypothesis*. Let $\sigma = w_1, w_2, \ldots$ be an infinite sequence, and $H_1, H_2, \ldots$ be the sequence of hypotheses produced by an inductive inference machine *IIM* when elements of σ are successively given to *IIM*. Then we say that *IIM* on *input* σ *converges to* H, if the sequence $H_1, H_2, \ldots$ of hypotheses is finite and ended by H, or there exists a positive integer n_0 such that $H_n = H$ for all $n \geqq n_0$. In this paper we assume that hypotheses produced by an inductive inference machine are indices of an indexed family. Discussion on more general setting of hypotheses in inductive inference can be found in [13].

A class of languages $\boldsymbol{L} = L_1, L_2, \ldots$ is said to be *inferable from positive* (or *complete*) *data* if there exists an inductive inference machine *IIM* such that, for any index i and any positive (or complete) presentation σ of L_i, *IIM* on input σ converges to H with $L_H = L_i$. We say that such *IIM identifies* L_i *in the limit* and *infers* the class $\boldsymbol{L}$ from positive (or complete) data.

Gold [9] showed that any indexed family of recursive languages is inferable from complete data. A class of languages is said to be *super-finite* if it contains all the finite languages and at least one infinite languages. Gold proved that any super-finite class is not inferable from positive data. Therefore, we know that inductive inference from positive data is strictly less powerful than that from complete data.

The limitation of the power of inductive inference from positive data mainly comes from overgeneralized hypotheses. A hypothesis H is said to be *overgeneralized* with respect to the target L_i if $L_i \subsetneq L_H$. All the positive examples of the target languages are also positive examples of any overgeneralized hypothesis. Does any inductive inference machine that does not produce any overgeneralized hypotheses identify the target? Although general answer to this question is "No" [2, 13], most known inductive inference machines take strategy to produce a minimal hypothesis.

Angluin [1, 2] characterized classes inferable from positive data by giving a necessary and sufficient condition and several sufficient conditions.

Let $\boldsymbol{L} = L_1, L_2, \ldots$ be a class of languages. A *finite tell-tale* of L_i is a finite subset T to L_i such that $T \subseteq L_j$ implies $\neg(L_j \subsetneq L_i)$ for any j.

Theorem 1 (Angluin [1, 2]). *A class* $\boldsymbol{L} = L_1, L_2, \ldots$ *is inferable from positive data if and only if there exists an effective procedure that enumerates the elements of a finite tell-tale of* L_i *for any* i.

The existence of finite tell-tale for each language in a class $\boldsymbol{L}$ is not sufficient for the inferability from positive data. However, if some language in $\boldsymbol{L}$ has no finite tell-tale, then $\boldsymbol{L}$ is not inferable from positive data. Theorem 1 does not clearly suggest concrete inductive inference machines, and its proof by Angluin presents a general inductive inference machine, which may not be easily understood. In many cases, however, we can use inductive inference machines which are intuitively understood. INFER_BY_MINL, we will introduce later, is one of such inductive inference machines.

A class $\boldsymbol{L}$ has *finite thickness* if $\#\{L \in \boldsymbol{L} \mid w \in L\}$ is finite for any word w. The notion of finite thickness is established by Angluin [1, 2] and named by Wright [24].

Theorem 2 (Angluin [1, 2]). *If a class has finite thickness then it is inferable from positive data.*

Here we give an informal proof of Theorem 2. Let $\boldsymbol{L} = L_1, L_2, \ldots$ be a class with finite thickness and consider the following procedure FTT that receives an index i and enumerates a subset of L_i. We assume that $w_1, w_2, \ldots$, is an effective enumeration of all the words over the alphabet and a function *Succ* gives the successor of a pair in some fixed enumeration of all the pairs of indices starting from $(1, 1)$.

procedure FTT;
input: An index i;
output: An enumeration of a subset of L_i;
method:
  begin
    $T_0 := \emptyset$;
    $n := 0$;
    $k_n := 1$;
    while $w_{k_n} \notin L_i$ **do** $k_n := k_n + 1$;
    output(w_{k_n});
    $n := n + 1$;
    $T_n := T_{n-1} \cup \{w_{k_{n-1}}\}$;
    repeat
      $(j_n, k_n) := (1, 1)$;
      while $T_n \notin L_{j_n}$ **or** $w_{k_n} \notin L_i - L_{j_n}$ **do** $(j_n, k_n) := Succ(j_n, k_n)$; $\quad (*)$
      output(w_{k_n});
      $n := n + 1$;
      $T_n := T_{n-1} \cup \{w_{k_{n-1}}\}$;
    forever
  end;

It is not hard to see that FTT enumerates a finite tell-tale of L_i. If the sequence of the outputs by FTT is infinite, then two infinite sequences $w_{k_0}, w_{k_1}, \ldots$ and $L_{j1}, L_{j2}, \ldots$ satisfy the condition $\{w_{k_0}, \ldots, w_{k_{n-1}}\} \subseteq L_{j_n}$ and $w_{k_n} \notin L_{j_n}$ for any $n = 1, 2, \ldots$. When such infinite sequences exist, we say that the class has *infinite elasticity*, which is discussed later. Note that words and languages in sequences are mutually different and $w_{k_0} \in L_{j_n}$ for all n. However, there exist only finitely many languages that contain w_{k_0} because the class has finite thickness. By contradiction, the output by FTT is a finite subset of L_i. If T_n is not a tell-tale of L_i, that is, $T_n \subseteq L_j \subsetneq L_i$ for some j, then there exists k such that $w_k \in L_i - L_j$. Therefore, the while loop $(*)$ eventually terminates and FTT outputs some word.

The most naive way to avoid overgeneralized hypothesis is to adopt the index i as a hypothesis such that L_i is a minimal language containing all the positive examples received so far. MINL calculation for a class $\boldsymbol{L} = L_1, L_2, \ldots$ is defined as follows:

$$\mathrm{MINL(S)} = \text{“Given nonempty finite set } S \text{ of words, find an index } i, \text{if exists, such that } S \subseteq L_i \text{ and for no index } j,\ S \subseteq L_j \subsetneq L_i\text{.”}$$

Then, the following procedure INFER_BY_MINL realizes an inductive inference machine.

procedure INFER_BY_MINL;
input: an infinite sequence $w_1, w_2, \ldots$ of words (a positive presentation);
output: an infinite sequence $H_1, H_2, \ldots$ of indices (hypotheses);
method:
  begin
    $H_0 :=$ “none”;
    $\mathbf{S} := \emptyset$;
    for $i := 1$ **to** ∞ **do**
      begin
        $S := S \cup \{w_i\}$;
        if $w_i \notin L_{H_{i-1}}$ **then**
          $H_i := \mathrm{MINL}(S)$
        else
          $H_i := H_{i-1}$;
        output H_i
      end
  end;

Theorem 3 (Angluin [1, 2]). *If a class* $\mathbf{L} = L_1, L_2, \ldots$ *has finite thickness and MINL for* $\mathbf{L}$ *is computable, then INFER_BY_MINL infers* $\mathbf{L}$ *form positive data.*

The proof of this theorem is similar to the discussion on the procedure FTT. INFER_BY_MINL produces only finitely many hypotheses because of the finite thickness of the class. If INFER_BY_MINL converges to H, then clearly L_H is equal to the target.

2.2. Patterns and their languages

Let Σ be a finite alphabet with $\#\Sigma \geqq 2$ and $X = \{x_1, x_2, \ldots\}$ be a countable alphabet disjoint from Σ. Elements in Σ are called *constants* and elements in X are called *variables*. A *pattern* is a non-empty word ever $\Sigma \cup X$. The set of variables appearing in a pattern p is denoted by $var(p)$. The number of occurrences of a variable x in a pattern p is denoted by $occ(x, p)$. A pattern p is said to be *m-variable* if $\#var(p) \leqq m$. A pattern p is said to be *proper* if $\#var(p) > 0$. A pattern containing no constant is said to be *pure*. A pattern p is said to be *regular* if $occ(x, p) = 1$ for each $x \in var(p)$. A regular pattern of the form $x_i w x_j$, where w is a constant word, is called a

substring pattern. For instance, ax_1bx_2a is regular but not one-variable, and ax_1bx_1a is one-variable but not regular, where a and b are constants. We denote the sets of all the patterns, m-variable patterns, proper patterns, pure patterns, regular patterns, and substring patterns by $\mathbf{P}, \mathbf{P}_m, \mathbf{Prop}, \mathbf{Pure}, \mathbf{RP}$, and $\mathbf{Psub}$, respectively.

A *substitution* is a homomorphism from pattern to pattern that maps every constant to itself. A substitution which maps some variables to the empty word is called an *erasing substitution*. When not stated, we assume that substitutions are non-erasing. For a pattern p and a substitution θ, we denote by $p\theta$ the image of p by θ. We use a set of replacements $\{x_1 := p_1, \ldots, x_k := p_k\}$ to represent a substitution which maps each variable x_i to a pattern p_i $(i = 1, \ldots, k)$ and every other symbol to itself. A *renaming of variables* is a substitution θ such that for any $x, y \in X$, (a) $x\theta \in X$, and (b) $x \neq y$ implies $x\theta \neq y\theta$. An equivalence relation $\equiv$ and subsumption relations $\preccurlyeq$ and $\preccurlyeq_\varepsilon$ on patterns are defined as follows:

$$p \equiv q \;\Leftrightarrow\; p = q\theta \text{ for some renaming of variables } \theta,$$

$$p \preccurlyeq q \;\Leftrightarrow\; p = q\theta \text{ for some non-erasing substitution } \theta,$$

$$p \preccurlyeq_\varepsilon q \;\Leftrightarrow\; p = q\theta \text{ for some possibly erasing substitution } \theta.$$

Precisely, $\preccurlyeq$ and $\preccurlyeq_\varepsilon$ are preorders but not partial orders, that is, they satisfy (a) reflexive law: $p \preccurlyeq p$, and (b) transitive law: $p \preccurlyeq q$ and $q \preccurlyeq r \Rightarrow p \preccurlyeq r$, but do not satisfy (c) antisymmetric law: $p \preccurlyeq q$ and $q \preccurlyeq p \Rightarrow p = q$. We can show that $p \equiv q \Leftrightarrow p \preccurlyeq q$ and $q \preccurlyeq p$. When we identify equivalent patterns with each other, however, we can think $\preccurlyeq$ as a partial order. We write $p \prec q$ if $p \preccurlyeq q$ but not $q \preccurlyeq p$. On the other hand, $\preccurlyeq_\varepsilon$ is not a partial order even when we identify equivalent patterns. For instance, $xy\{y := \varepsilon\} = x$ and $x\{x := xy\} = xy$, therefore $xy \preccurlyeq_\varepsilon x \preccurlyeq_\varepsilon xy$, but xy is not equivalent to x.

The *language of a pattern* p, denoted by $L(p)$, is the set $\{w \in \Sigma^+ \mid w \preccurlyeq p\}$. For a set S of patterns, $L(S)$ denotes the class of languages defined by $\boldsymbol{L}(S) = \{L(p) \mid p \in S\}$. When erasing substitutions are allowed, pattern languages are called extended languages [18]. The *extended language* of a pattern p, denoted by $L_\varepsilon(p)$, is the set $\{w \in \Sigma^+ \mid w \preccurlyeq_\varepsilon p\}$. $\boldsymbol{L}_\varepsilon(\boldsymbol{S})$ denotes the class $\{L_\varepsilon(p) \mid p \in S\}$.

Lemma 4 (Angluin [1]). *For any patterns p and q, $p \preccurlyeq q$ implies $L(p) \subseteq L(q)$, but the converse does not hold in general.*

Here we give examples showing that the subsumption relation between patterns does not coincide with the inclusion relation between their languages. Let $\Sigma = \{a, b\}$, $X = \{x_1, x_2, \ldots\}$.

(1) If $p = ax_1bax_1x_1b$, $q = x_1x_1x_2$, then $\neg(p \preccurlyeq q)$ but $L(p) \subseteq L(q)$ [1].
(2) If $p = aax_1bb$ and $q = x_1abx_2$, then $\neg(p \preccurlyeq q)$ but $L(p) \subseteq L(q)$ [15].
(3) If $p = x_1abx_2ax_3$ and $q = x_1ax_2bax_3$, then $\neg(p \preccurlyeq_\varepsilon q)$ and $\neg(q \preccurlyeq_\varepsilon p)$, but $L_\varepsilon(p) = L_\varepsilon(q)$ [18].

Note that patterns in (2) and (3) are regular while patterns in (1) are not regular. The above examples are the case only if $\#\Sigma = 2$. When Σ contains three or more constant symbols, p and q in (1) can be modified to keep the condition $\neg(p \preccurlyeq q)$ but $L(p) \subseteq L(q)$ [1]. For example, $p = ax_1bax_1x_1cax_1bax_1x_1x_1c$ and $q = x_1x_1x_2$ keep the condition for $\Sigma = \{a, b, c\}$. On the other hand, for any regular patterns p and q, we can show that $p \preccurlyeq q \Leftrightarrow L(p) \subseteq L(q)$ and $p \preccurlyeq_\varepsilon q \Leftrightarrow L_\varepsilon(p) \subseteq L_\varepsilon(q)$ when $\#\Sigma \geqq 3$ [15, 18]. In general the inclusion problem of pattern languages is undecidable [11]. The inclusion problem of pure pattern languages is decidable [11].

Lemma 5 (Angluin [1]). *For any patterns p and q, $p \preccurlyeq q$ implies $|p| \geqq |q|$.*

By Lemma 5 and the fact that there are only finitely many inequivalent patterns of length at most l, we can show that $\boldsymbol{L}(\boldsymbol{P})$ has finite thickness.

Lemma 6 (Angluin [1]). *The class $\boldsymbol{L}(\boldsymbol{P})$ of pattern languages has finite thickness.*

Theorem 7 (Angluin [1, 2]). *The class $\boldsymbol{L}(\boldsymbol{P})$ of pattern languages is inferable from positive data.*

2.3. Inductive inference of unions

Inductive inference of unions is a task to identify a union of several languages from data without any information about its origin. Here, we show that inductive inference from positive data is possible for the class of unions of at most k pattern languages, according to Wright [24]. Shinohara [19] proved a special case $k = 2$. Wright [24] extended the result to the general case. Arimura et al. [6, 7] showed an efficient inference algorithm for unions of at most k regular pattern languages.

Let $\boldsymbol{L} = L_1, L_2, \ldots$ be a class of languages and $I = \{1, 2, \ldots\}$ be the set of indices. For each $k \geqq 1$, we define $I^k = \{S \mid S \subseteq I \text{ and } 1 \leqq \#S \leqq k\}$, $L_S = \bigcup_{i \in S} L_i$, and the class of unions $\boldsymbol{L}^k = \{L_S \mid S \in I^k\}$. For a set S of patterns we also write $L(S)$ instead of L_S. Similarly, we define $I^* = \{S \mid S \subseteq I \text{ and } 1 \leqq \#S < \infty\}$ and the class of unbounded unions by $\boldsymbol{L}^* = \{L_S \mid S \in I^*\}$. We also use notations such as $\boldsymbol{P}^k$ and $\boldsymbol{L}(\boldsymbol{P}^k)$ to denote the class of sets of at most k patterns and the class of unions of at most k pattern languages, respectively. It is natural to ask whether $\boldsymbol{L}^k$ is inferable from positive data whenever so is $\boldsymbol{L}$. It is not hard to see that the answer to this question is "not in general". Wright introduced a notion of finite elasticity [24, 14], which is an extension of finite thickness and sufficient for inferability from positive data. He showed that the property of finite elasticity is closed under union.

A class $\boldsymbol{L}$ has *infinite elasticity* if there exist two infinite sequences

$$w_0, w_1, w_2, \ldots \quad \text{and} \quad L_1, L_2, L_3, \ldots,$$

where $w_i \in \Sigma^*(i = 0, 1, \ldots)$ and $L_j \in \boldsymbol{L}(j = 1, 2, \ldots)$, such that

$$\{w_0, w_1, \ldots, w_{i-1}\} \subseteq L_i \quad \text{but } w_i \notin L_i$$

for any $i = 1, 2, \ldots$. $\boldsymbol{L}$ has *infinite elasticity* if $\boldsymbol{L}$ does not have infinite elasticity. Note that words $w_0, w_1, w_2, \ldots$ and languages $L_1, L_2, L_3, \ldots$ in the definition should be mutually different. Then, it is easy to see that any class with finite thickness always has finite elasticity.

Lemma 8 (Wright [24]). *If $\boldsymbol{L}$ has finite elasticity then $\boldsymbol{L}^k$ also has finite elasticity for any $k \geqq 1$.*

The following theorem for classes with finite elasticity can be proved in a similar way to Theorem 2 by using the procedure FTT. Furthermore, if MINL for the class with finite elasticity is computable, then we can use procedure INFER_BY_MINL as inductive inference machine.

Theorem 9 (Wright [24]). *If $\boldsymbol{L}$ has finite elasticity then $\boldsymbol{L}$ is inferable from positive data.*

From Lemmas 6, 8 and Theorem 9 we have the following.

Theorem 10 (Wright [24]). *The class $\boldsymbol{L}(\boldsymbol{P}^k)$ of unions of at most k pattern languages is inferable from positive data for any $k \geqq 1$.*

Here we should note that the class $\boldsymbol{L}^*$ of unions of unbounded number of languages is out of the scope of Wright's results.

2.4. Extended pattern languages

The inferability of extended pattern languages from positive data is still open in general. Lemmas 5 and 6 are not the case for extended pattern languages. In fact, the class of extended pattern languages does not have finite thickness, because infinitely many patterns $x, xx, xxx, \ldots$ define mutually different extended languages that contain the empty word ε. For some subclasses of extended pattern languages, the inferability from positive data is known. When we consider extended languages, any consecutive occurrence of variables in regular patterns can be replaced with a single variable. For example, $L_\varepsilon(x_1 a x_2 x_3 x_4 b x_5) = L_\varepsilon(x_1 a x_2 b x_5)$. Thus, the class of extended regular pattern languages has finite thickness.

Lemma 11 (Shinohara [18]). *For any patterns p and q, $p \preccurlyeq_\varepsilon q$ implies $L_\varepsilon(p) \subseteq L_\varepsilon(q)$, but the converse does not hold in general.*

Lemma 12 (Shinohara [18]). *The class $\boldsymbol{L}_\varepsilon(\boldsymbol{RP})$ of extended regular pattern languages has finite thickness.*

Theorem 13 (Shinohara [18]). *The class $\boldsymbol{L}_\varepsilon(\boldsymbol{RP})$ of extended regular pattern languages in inferable from positive data.*

Any extended language of a pattern p can be defined as a union of non-extended pattern languages. In fact, $L_\varepsilon(p) = \bigcup_{q \in \Pi(p)} L(q)$, where $\Pi(p)$ is the set of patterns obtained by erasing some of the variables in p. For example, $L_\varepsilon(axby) = L(axby) \cup L(aby) \cup L(axb) \cup L(ab)$. If a pattern p contains m variables, then the extended language $L_\varepsilon(p)$ is defined as a union of at most 2^m non-extended pattern languages. Thus, we have the following result.

Corollary 14 (Wright [24]). *The class $\boldsymbol{L}_\varepsilon(\boldsymbol{P}_m)$ of extended m-variable pattern languages is inferable from positive data for any $m \geqq 1$.*

As for the elasticity of $\boldsymbol{L}_\varepsilon(\boldsymbol{P})$, we can show the following theorem.

Theorem 15. *The class $\boldsymbol{L}_\varepsilon(\boldsymbol{Pure})$ of extended pattern languages has infinite elasticity.*

Proof. Let $\Sigma = \{a, b\}$, $X = \{x_1, x_2, \ldots, y_1, y_2, \ldots\}$, and $Pr(n)$ denote the nth prime number. We define words by $w_i = ab^{Pr(i+1)}a$ for $i = 0, 1, \ldots$, and patterns by $p_i = x_i y_i^{Pr(i)} x_i$ and $q_i = p_1 \cdots p_i$ for $i = 1, 2, 3, \ldots,$. If $w_i \in L_\varepsilon(q_j)$, then there exists a substitution θ such that $w_i = qj\theta$. Since a appears exactly twice in w_i and $y_m (2 \leqq m \leqq j)$ appears at least three time in q_j, there uniquely exists k such that $1 \leqq k \leqq j$, $x_k\theta = a, y_k^{Pr(k)\theta} = b^{Pr(i+1)}$, and $x_{k'}\theta = \varepsilon$, and $y_{k'}\theta = \varepsilon$ for any $k' \neq k$. Since $Pr(k)$ and $Pr(i+1)$ are prime numbers, $k = i+1$. Therefore, $i < j$. Thus, we have $\{w_0, \ldots, w_{i-1}\} \subseteq L_\varepsilon(q_i)$ but $w_i \notin L_\varepsilon(q_i)$, which show the infinite elasticity. □

3. Proper pattern languages

In this section, we investigate the inferability of unbounded unions of proper pattern languages. First we consider substring patterns, which can be considered as the simplest proper patterns.

3.1. Substring pattern languages

Substring patterns are very simple proper patterns. Note that there are infinitely many substring patterns and the inferability of their unbounded unions is non-trivial. Furthermore, we can show that the class of unbounded unions has infinite elasticity.

Theorem 16. *The class $\boldsymbol{L}_{[\varepsilon]}(\boldsymbol{Psub}^*)$ of unbounded unions of (extended) substring pattern languages has infinite elasticity.*

Proof. Let $\Sigma = \{a, b, \ldots\}$, $p_i = x_1 a s_i a x_2$, where x_1 and x_2 are variables and $s_0 = \varepsilon, s_1, \ldots$ is an enumeration of all the words in $(\Sigma - \{a\})^*$. Let $L_i = L(p_0) \cup \cdots \cup L(p_i)$ for each $i = 0, 1, \ldots$.

Since $aas_iaa \in L(p_j) \Leftrightarrow i = j$, we have infinite sequence $L_0 \subsetneq L_1 \subsetneq \cdots$. Therefore, the class has finite elasticity. For extended pattern languages, similar proof works. □

Theorem 17. *The class* $\boldsymbol{L}_{[\varepsilon]}(\boldsymbol{Psub})^*$ *of unbounded unions of* (*extended*) *substring pattern languages is inferable from positive data.*

Proof. We show that any union of substring pattern languages has a finite tell-tale. Let $A=\{x_1w_1x_2,\ldots,x_1w_mx_2\}$ be any finite set of substring patterns. Clearly, $T=\{aw_1a,\ldots,aw_ma\}$ is a finite subset of $L(A)$. Let $B=\{x_1v_1x_2,\ldots,x_1v_nx_2\}$ and $T\subseteq L(B)$. Then, for any $i=1,\ldots,m$, there exists j such that $aw_ia\in L(x_1v_jx_2)$, $1\leqq j\leqq n$, which implies that v_j is a substring of w_i, that is, $w_i=u_1v_ju_2$ for some $u_1,u_2\in\Sigma^*$. Therefore, $x_1w_ix_2\preccurlyeq x_1v_jx_2$, and by Lemma 4 $L(x_1w_ix_2)\subseteq L(x_1v_jx_2)$. Thus, $L(A)\subseteq L(B)$. This shows that T is a finite tell-tale of $L(A)$. By Theorem 1, the proof is done. For extended languages, we can similarly prove that $T_\varepsilon=\{w_1,\ldots,w_m\}$ is a finite tell-tale of $L_\varepsilon(A)$. □

3.2. Proper pattern languages

Here, we show that the class of unbounded unions of proper pattern languages is not inferable from positive data, even is we restrict patterns to be regular or one-variable as well as proper. The following lemma is a generalization of the Gold's theorem on super-finite classes.

Lemma 18. *If there exists an infinite sequence* $L_0\subsetneq L_1\subsetneq\cdots$ *in a class* $\boldsymbol{L}$ *of languages such that* $L_\infty=\lim_{n\to\infty}L_0\cup\cdots\cup L_n\in\boldsymbol{L}$, *then* $\boldsymbol{L}$ *is not inferable from positive data.*

Proof. Let $L_0\subsetneq L_1\subsetneq\cdots$ be an infinite sequence of languages in a class $\boldsymbol{L}$ and $L_\infty=\lim_{n\to\infty}L_0\cup\cdots\cup L_n\in\boldsymbol{L}$. Let T be any finite subset of L_∞. Then, $T\subseteq L_i\subsetneq L_\infty$ for some i. Therefore, T cannot be a finite tell-tale of L_∞. By Theorem 1, $\boldsymbol{L}$ is not inferable from positive data. □

Theorem 19. *The class* $\boldsymbol{L}_{[\varepsilon]}(\boldsymbol{RProp}^*)$ *of unbounded unions of* (*extended*) *proper regular pattern languages is not inferable from positive data.*

Proof. Let patterns $p_0,p_1,\ldots$ and languages $L_0\subsetneq L_1\subsetneq\cdots$ be the same as in the proof of Theorem 16. Let $L_\infty=\lim_{n\to\infty}L_n$ and $L=L(x_1aax_2)\cup L(x_1ax_2ax_3)$. Since $p_0=x_1aax_2$ and $p_i\preccurlyeq x_1ax_2ax_3$ for any $i\geqq 1$, $L(p_i)\subseteq L$ for any $i\geqq 0$, by Lemma 4. Therefore, $L_\infty\subseteq L$. Let $w\in L$. If $w\in L(x_1aax_2)$, then $w\in L_\infty$. If $w\in L(x_1ax_2ax_3)$, then $w=w_1aw_2aw_3$ for some $w_1,w_2,w_3\in\Sigma^+$. In case $w_2=aw_2'$ for some $w_2'\in\Sigma^*$, $w=w_1aaw_2'aw_3\in L(x_1aax_2)$. In case $w_2=cw_2'$ for some $c\in\Sigma-\{a\}$ and $w_2'\in\Sigma^*$, $w=w_1acw_2'aw_3\in L(x_1as_iax_2)$ for some $i\geqq 1$. Therefore, $L\subseteq L_\infty$. Thus, $L_\infty=L$. For extended languages we can similarly show $L_\infty=L_\varepsilon(x_1ax_2ax_3)$. From Lemma 18 we have the theorem. □

Here we should note that all the patterns $p_0,p_1,\ldots$ in the above proof are substring patterns but the pattern $x_1ax_2ax_3$ is not a substring pattern.

For one-variable pattern languages, we have the following negative result.

Theorem 20. *The class* $\boldsymbol{L}_{[\varepsilon]}(\boldsymbol{Prop}_1^*)$ *of unbounded unions of* (*extended*) *one-variable proper pattern languages is not inferable from positive data.*

Proof. For simplicity, we assume $\Sigma = \{a, b\}$ and $x \in X$. We define a series $L_0, L_1, \ldots$ of unions of one-variable pattern languages and a union L_∞ by

$$L_0 = L(xx) \cup L(axx) \cup L(xb) \cup L(bx),$$

$$L_n = L_{n-1} \cup L(a^n bx) \text{ for } n \geqq 1, \text{ and}$$

$$L = L(ax) \cup L(bx).$$

Note that L is the set of all the words of length two or more. Let w be any word of length at least two. If w contains no b, that is, $w = a^i$ for some $i \geqq 2$, then $w \in L(xx) \cup L(axx)$. If w contains at least one b, then $w = a^i bu$ for some $i \geqq 0$ and some $u \in \Sigma^*$. In case $i = 0$ or $|u| = 0, w \in L(bx)$ or $w \in L(xb)$, respectively. Otherwise, $w \in L(a^i bx)$. Therefore, L_n contains all words of length $2, 3, \ldots, n+2$. Clearly, $a^{n+1}ba \notin L_n$ for any $n = 0, 1, \ldots$. Thus, we have an infinite sequence $L_1 \subsetneq L_2 \subsetneq \cdots$ in the class of unions of one-variable proper pattern languages such that $L = \lim_{n\to\infty} L_n$. By Lemma 18, the class is not inferable from positive data.

We can easily modify the proof for general case that Σ contains three or more symbols.

The similar argument to the above is also valid for extended languages if we define $L = L_\varepsilon(x) = \Sigma^*$. □

4. Regular patterns with bounded constant segments

In this section, we consider subclasses of regular patterns by bounding the length of consecutive occurrences of constant symbols and show the inferability from positive data of unions of their languages. For example, a regular pattern "$x_1abx_2abcx_3$" has two *constant segments* "*ab*" and "*abc*" of length 2 and 3, respectively. To show the inferability of unions from positive data, we use Higman's Theorem.

4.1. Infinite anti-chain and Higman's theorem

Let $\geqq$ be a preorder over a set A. An *infinite anti-chain* over A with respect to $\geqq$ is an infinite sequence $a_0, a_1, \ldots, a_i, \ldots$ such that $a_i \in A$ for any $i \geqq 0$, and $i \neq j$ implies a_i and a_j are incomparable, that is, neither $a_i \geqq a_j$ nor $a_j \geqq a_i$.

Let Σ be a finite alphabet. The *subsequence relation* is a binary relation $\geqslant_s$ over Σ^* defined by for any sequence $u = a_1 \cdots a_m$ and $v = b_1 \cdots b_n$, $u \geqslant_s v$ if and only if u is a subsequence of v, that is, $a_1 = b_{i_1}, \ldots, a_m = b_{i_m}$ for some $1 \leqq i_1 < \cdots < i_m \leqq n$. The following lemma is known as *Higman's Theorem* in combinatorics on strings. For a proof of lemma, see Kruskal [12].

Lemma 21 (Higman's Theorem). *Let Σ be a finite alphabet. Then, there is no infinite anti-chain of strings in Σ^* with respect to the subsequence relation $\succcurlyeq_s$.*

4.2. Infinite anti-chain and inferability of unions from positive data

Lemma 22. *Let $\boldsymbol{L}$ be a class with finite thickness. If $\boldsymbol{L}$ has no infinite anti-chain with respect to the set inclusion $\supseteq$, then the class $\boldsymbol{L}^*$ of unions is inferable from positive data.*

Proof. Let $\boldsymbol{L}$ be a class with finite thickness and no infinite anti-chain with respect to the set inclusion $\supseteq$.

Assume the class $\boldsymbol{L}^*$ of unions has infinite elasticity. Then there exist two infinite sequences of words and finite set of languages,

$$w_0, w_1, w_2, \ldots \quad \text{and} \quad U_1, U_2, U_3, \ldots$$

such the $\{w_0, \ldots, w_{i-1}\} \subseteq \bigcup_{L \in U_i} L$ but $w_i \notin \bigcup_{L \in U_i} L$ for any i. Here we should note that $w_0 \in \bigcup_{L \in U_i} L$ for any i. Since $\boldsymbol{L}$ has finite thickness, there exist only finitely many languages containing w_0. Therefore, there exists a language L_1 such that $w_0 \in L_1$ and L_1 appears in infinitely many U_i's. Let $U_{i_1}, U_{i_2}, U_{i_3}, \ldots$ be an infinite subsequence of $U_1, U_2, U_3, \ldots$ such that $L_1 \in U_{i_k}$ for any k. Then two infinite sequences

$$w_0, w_{i_1}, w_{i_2}, \ldots \quad \text{and} \quad U_{i_1}, U_{i_2}, U_{i_3}, \ldots$$

also show the infinite elasticity of L^*. Note that $w_{i_1} \notin L_1$. By renumbering the above sequences, we can assume that the two infinite sequences

$$w_0, w_1, w_2, \ldots \quad \text{and} \quad U_1, U_2, U_3, \ldots,$$

that show the infinite elasticity, satisfy

$$U_1 = \{L_1\}, w_0 \in L_1, L_1 \in U_i, \quad \text{and}$$
$$\{w_0, \ldots, w_{i-1}\} \subseteq \bigcup_{L \in U_i} L \quad \text{but } w_i \notin \bigcup_{L \in U_i} L \quad \text{for any } i.$$

Similarly, we can take a language L_2 that contains w_1 and appears in infinitely many U_i's. Thus, we have infinite sequences

$$w_0, w_1, w_2, \ldots \quad \text{and} \quad U_1 = \{L_1\},\ U_2 = \{L_1, L_2\},\ U_3 = \{L_1, L_2, L_3\}, \ldots$$

that show the infinite elasticity. There may exist some $j > 1$ such that $w_0 \in L_j$. The number of such j's is finite because the class has finite thickness. Therefore, there exists an infinite sequence $j_0 < j_1 < j_2 \ldots$ such that $w_{j_{k-1}} \in L_{j_k}$ if and only if $k = k'$. Clearly, $L_{j_1}, L_{j_2}, L_{j_3}, \ldots$ is an infinite anti-chain over $\boldsymbol{L}$ with respect to the inclusion relation $\supseteq$. By contradiction, $\boldsymbol{L}^*$ does not have infinite elasticity.

By Theorem 9, $\boldsymbol{L}^*$ is inferable from positive data. □

4.3. Inferability of unions of regular pattern languages

Let $\boldsymbol{RP}_l$ be the set of regular patterns of the form $p = w_0 x_1 w_1 \cdots w_{n-1} x_n w_n$ where $n \geqq 0$, $x_0, \ldots, x_n$ are mutually distinct variables, and $w_0, \ldots, w_n \in \Sigma^{\leqq l}$. We call each w_i $(1 \leqq i \leqq n)$ a *constant segment* in p. We say that p *begins* (*ends*) *with a variable* if $w_0 = \varepsilon$ ($w_n = \varepsilon$). By $x(\boldsymbol{RP}_l)y$, we denote the subset of $\boldsymbol{RP}_l$ whose patterns both begin and end with a variable.

Lemma 23. *Let $\geqq$ and $\geqq'$ be preorders over a set S such that $p \geqq q$ implies $p \geqq' q$ for any $p, q \in S$. If there is no infinite anti-chain over S with respect to $\geqq$, then there is no infinite anti-chain over S with respect to $\geqq'$.*

Lemma 24. *Let Σ be finite alphabet, and $l \geqq 0$. Then, there is no infinite anti-chain over $x(\boldsymbol{RP}_l)y$ with respect to $\succcurlyeq$.*

Proof. We define $p \succcurlyeq' q$ if $p = x_0 u_1 x_1 \cdots u_m x_m$, $q = y_0 v_1 y_1 \cdots v_n y_n$, and $u_1, \ldots, u_m$ is a subsequence of $v_1, \ldots, v_n$, where $x_0, \ldots, x_m, y_0, \ldots, y_n$ are variables and $u_1, \ldots, u_m$, $v_1, \ldots, v_n \in \Sigma^{\leqq l}$. The relation $\succcurlyeq'$ over $x(\boldsymbol{RP}_l)y$ is isomorphic to $\succcurlyeq_s$ over $(\Sigma^{\leqq l})^*$. Clearly, $p \succcurlyeq' q$ implies $p \succcurlyeq q$ for any $p, q, \in x(\boldsymbol{RP}_l)y$. From Lemmas 21 and 23, the result follows. □

Lemma 25. *Let Σ be a finite alphabet, and $l \geqq 0$. Then, there is no infinite anti-chain over $\boldsymbol{RP}_l$ with respect to $\succcurlyeq$.*

Proof. Suppose that there exists an infinite anti-chain $\tau = p_0, p_1, \ldots$ over $\boldsymbol{RP}_l$ with respect to $\succcurlyeq$. Clearly, for each $i = 0, 1, \ldots$, $p_i = u_i q_i v_i$, for some $u_i \in \Sigma^{\leqq l}$, $q_i \in x(\boldsymbol{RP}_l)y$, and $v_i \in \Sigma^{\leqq l}$. Since $\Sigma^{\leqq l}$ is finite, there exist $u, v, \in \Sigma^{\leqq l}$ such that $u_i = u$ and $v_i = v$ for infinitely many i. Let $\tau' = uq_{i_0}v, uq_{i_1}v, \ldots$ be an infinite subsequence of τ. Then, $q_{i_0}, q_{i_1}, \ldots$ is an infinite anti-chain over $x(\boldsymbol{RP}_l)y$ with respect to $\succcurlyeq$. This contradicts Lemma 24. □

Lemma 26. *Let Σ be a finite alphabet, and $l \geqq 0$. Then, there is no infinite anti-chain over $\boldsymbol{RP}_l$ with respect to $\succcurlyeq_\varepsilon$.*

Proof. For any $p, q \in \boldsymbol{RP}_l$, $p \succcurlyeq q$ implies $p \succcurlyeq_\varepsilon q$. From Lemmas 23 and 25, the result immediately follows. This result does not depend on the existence of the empty pattern ε in $\boldsymbol{RP}_l$. □

Theorem 27. *For any $l > 0$, the class $\boldsymbol{L}(\boldsymbol{RP}_l^*)$ of unions of languages defined by regular patterns with constant segments of length at most l is inferable from positive data.*

Proof. Assume that there exists an infinite anti-chain over $\boldsymbol{L}(\boldsymbol{RP}_l)$ with respect to the set inclusion $\supseteq$. Then there exists an infinite sequence $\tau = p_1, p_2, \ldots$ of patterns in

$\boldsymbol{RP}_l$ such that $i \neq j$ implies $L(p_i) \not\supseteq L(p_j)$ and $L(p_j) \not\supseteq L(p_i)$. From Lemma 4, for any patterns p and q, $p \gtrdot q$ implies $L(p) \supseteq L(p)$. Therefore, τ is an infinite anti-chain over $\boldsymbol{RP}_l$ with respect to $\gtrdot$. However, Lemma 25 says that there is no such infinite anti-chain. By contradiction, there is no infinite anti-chain over $\boldsymbol{L}(\boldsymbol{RP}_l)$ with respect to $\supseteq$. Recall that $\boldsymbol{L}(\boldsymbol{P})$ has finite thickness. Clearly, $\boldsymbol{L}(\boldsymbol{RP}_l)$ also has finite thickness. Finally, from Lemma 22, the class $\boldsymbol{L}(\boldsymbol{RP}_l^*)$ of unions is inferable from positive data. □

Theorem 28. *For any $l \geqq 1$, the class $\boldsymbol{L}_\varepsilon(\boldsymbol{RP}_l^*)$ of unions of extended languages defined by regular patterns with constant segments of length at most l is inferable from positive data.*

Proof. Similar to the proof of Theorem 27. Use Lemma 26 instead of Lemma 25. □

5. Concluding remarks

In this paper, we have discussed unbounded unions of languages from the viewpoint of inferability from positive data. Unfortunately, the problem for proper pattern languages, which directly motivated us to attack unbounded unions, has been negatively solved. We have also seen several non-trivial classes of unbounded unions of pattern languages to be inferable from positive data. As for the inferability of extended pattern languages, we have shown that the class has infinite elasticity even if we restrict patterns to be pure. From the relation between extended pattern languages and unions of non-extended pattern languages, the class $\boldsymbol{L}(\boldsymbol{Pure}^*)$ of unbounded unions of pure pattern languages also has infinite elasticity. Therefore, the inferability of $\boldsymbol{L}(\boldsymbol{Pure}^*)$ and $\boldsymbol{L}_\varepsilon(\boldsymbol{Pure})$ might be a challenging problem before attacking $\boldsymbol{L}_\varepsilon(\boldsymbol{P})$. Table 1 summarizes the inferability of pattern languages and unions.

As a future work, we should consider efficient inference algorithms for unions. The class of unions of extended languages of restricted regular patterns, that starts and ends with variable and contain constant segments of length at most one, is almost trivially inferable from positive data because their languages are defined by using the subsequence relation. For example, the extended language $L_\varepsilon(xaybz)$ is the set of strings containing a and b in this order, that is, *supersequences* of ab. Therefore, we call such languages *subsequence languages*. To infer a union of subsequence languages from positive data, collecting examples and forgetting unnecessary examples, which are subsequences of examples collected so far, is enough to compule MINL and INFER_BY_MINL can successfully infer the class. However, in case of non-extended languages or longer constant segments, we should investigate concrete inference algorithms.

The terms in the first-order logic can be considered as patterns with tree structure, and therefore, they are also called *tree patterns* the languages of which are set of all the ground instances of them. Arimura et al. [3] investigate inductive inference of unions of at most k tree pattern languages from positve data, where *k-mmg*

Table 1
Inferability of Pattern Languages and Unions

Category	Classes	Inferability	Elasticity	Thickness
Pattern languages	$\boldsymbol{L}(\boldsymbol{P})$[a]	Yes	Finite	Finite
Extended pattern languages	$\boldsymbol{L}_{\varepsilon}(\boldsymbol{P})$	Open	Infinite[d]	Infinite
	$\boldsymbol{L}_{\varepsilon}(\boldsymbol{RP})$[b]	Yes	Finite	Finite
	$\boldsymbol{L}_{\varepsilon}(\boldsymbol{P}_m)$[c]	Yes	Finite	Infinite
	$\boldsymbol{L}_{\varepsilon}(\boldsymbol{Pure})$	Open	Infinite[d]	Infinite
Bounded unions	$\boldsymbol{L}_{\varepsilon}(\boldsymbol{P}^k)$[c]	Yes	Finite	Infinite
Unbounded unions[(4)]	$\boldsymbol{L}_{[\varepsilon]}(\boldsymbol{Prop}^*)$	No	Infinite	Infinite
	$\boldsymbol{L}_{[\varepsilon]}(\boldsymbol{Prop}_1^*)$	No	Infinite	Infinite
	$\boldsymbol{L}_{[\varepsilon]}(\boldsymbol{RProp}^*)$	No	Infinite	Infinite
	$\boldsymbol{L}_{[\varepsilon]}(\boldsymbol{PSub}^*)$	Yes	Infinite	Infinite
	$\boldsymbol{L}_{[\varepsilon]}(\boldsymbol{RP}_l^*)$	Yes	Finite	Infinite
	$\boldsymbol{L}(\boldsymbol{Pure}^*)$	Open	Infinite	Infinite

Note: (1) Angluin [1], (2) Shinohara [18], (3) Wright [24], (4) This paper.

(k-minimal multiple generalization) algorithm is used as MINL calculation. Arikawa et al. [8] proposed *elementary formal systems* (EFS's for short) [22], which are finite sets of definite clauses on patterns, and considered as a kind of logic programs, as a unifying framework for language learning. Shinohara [20, 21] extended the results on unions of patterns languages by Wright [24] to languages defined by EFSs consisting of at most k definite clauses. Discussions in the present paper suggests that there may exist classes of unbounded unions of tree pattern languages and languages of EFS's with unbounded number of definite clauses inferable from positive data.

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